

# 11.4

1) An equation of an ellipse is given.

$$\frac{(x-2)^2}{16} + \frac{(y-3)^2}{4} = 1$$

Find center, vertices, foci

x-major

Center:  $(2, 3)$

$$c^2 = a^2 - b^2$$

$$c^2 = 16 - 4$$

$$c^2 = 12$$

$$c = 2\sqrt{3}$$

Focus:  $(2 - 2\sqrt{3}, 3)$

Focus:  $(2 + 2\sqrt{3}, 3)$

$$a^2 = 16$$

$$a = 4$$

Vertex:  $(2 - 4, 3)$

$(-2, 3)$

Vertex:  $(2 + 4, 3)$

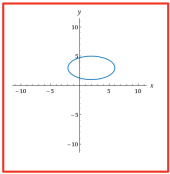
$(6, 3)$

lengths of major and minor axes

major axis:  $2a = 2 \cdot 4 = 8 \text{ un}$

minor axis:  $2b = 2 \cdot 2 = 4 \text{ un}$

graph



2) An equation of a parabola is given.

$$(x-4)^2 = 4(y+4)$$

Find the vertex, focus, and directrix

Vertex:  $(4, -4)$

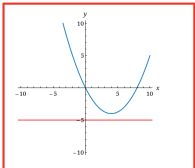
Focus:  $(4, -4 + 1)$

$(4, -3)$

Directrix:  $y = -4 - 1$

$y = -5$

Graph



3) An equation of a hyperbola is given.

$$\frac{(x+3)^2}{9} - \frac{(y-4)^2}{16} = 1$$

Find the center, vertices, foci, and asymptotes.

Center:  $(-3, 4)$

Vertex:  $(-3 - 3, 4)$

$(-6, 4)$

Vertex:  $(-3 + 3, 4)$

$(0, 4)$

$$c^2 = a^2 + b^2$$

$$c^2 = 9 + 16$$

$$c^2 = 25$$

$$c = 5$$

Focus:  $(-3 - 5, 4)$

$(-8, 4)$

Focus:  $(-3 + 5, 4)$

$(2, 4)$

Asymptotes:

$$b^2 = 16$$

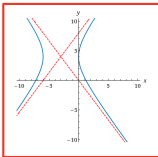
$$b = 4$$

$$a^2 = 9$$

$$a = 3$$

$$y - 4 = \pm \frac{4}{3}(x + 3)$$

Graph



4) An equation of an ellipse is given.

$$x^2 + \frac{(y+4)^2}{16} = 1$$

Find center, vertices, foci

y-major

center:  $(0, -4)$

$$c^2 = a^2 - b^2$$

$$c^2 = 16 - 1$$

$$c^2 = 15$$

$$c = \sqrt{15}$$

Focus:  $(0, -4 - \sqrt{15})$

Focus:  $(0, -4 + \sqrt{15})$

Vertex:  $(0, -4 - 4)$

$(0, -8)$

Vertex:  $(0, -4 + 4)$

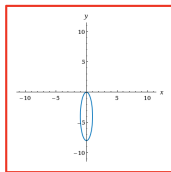
$(0, 0)$

lengths of major and minor axes

major axis:  $2a = 2 \cdot 4 = 8 \text{ un}$

minor axis:  $2b = 2 \cdot 1 = 2 \text{ un}$

Graph



5) An equation of a parabola is given.

$$x^2 + 6x - 20y + 49 = 0$$

Find the vertex, focus, and directrix

$$x^2 + 6x = 20y - 49$$

$$x^2 + 6x + 9 = 20y - 49 + 9$$

$$(x+3)^2 = 20(y-2)$$

Vertex:  $(-3, 2)$

$$4p = 20$$

$$p = 5$$

opens up

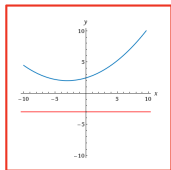
Focus:  $(-3, 2+5)$

$(-3, 7)$

Directrix:  $y = 2 - 5$

$y = -3$

Graph



6) An equation of a hyperbola is given.

$$25x^2 - 16y^2 - 96y = 544$$

Find the center, vertices, foci, and asymptotes.

$$25x^2 - 16(y^2 + 6y + 9 - 9) = 544$$

$$25x^2 - 16(y+3)^2 - 144 = 544$$

$$25x^2 - 16(y+3)^2 + 144 = 544$$

$$25x^2 - 16(y+3)^2 = 400$$

$$\frac{25x^2}{400} - \frac{16(y+3)^2}{400} = 1$$

$$\frac{x^2}{16} - \frac{(y+3)^2}{25} = 1$$

x-major

center:  $(0, -3)$

$$a^2 = 16$$

$$a = 4$$

$$\text{vertex: } (0-4, -3)$$

$$(-4, -3)$$

$$\text{vertex: } (0+4, -3)$$

$$(4, -3)$$

$$c^2 = a^2 + b^2$$

$$c^2 = 16 + 25$$

$$c^2 = 41$$

$$c = \sqrt{41}$$

$$\text{Focus: } (0-\sqrt{41}, -3)$$

$$(-\sqrt{41}, -3)$$

$$\text{Focus: } (0+\sqrt{41}, -3)$$

$$(\sqrt{41}, -3)$$

Asymptotes:

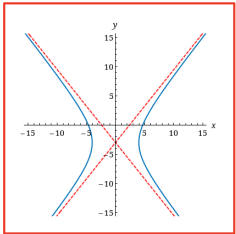
$$a = 4$$

$$b^2 = 25$$

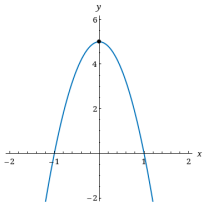
$$b = 5$$

$$y+3 = \pm \frac{5}{4}x$$

Graph



7) Find an equation for the conic whose graph is shown



$$\text{Vertex: } (0, 5)$$

$$x\text{-ints: } (-1, 0), (1, 0)$$

$$y = a(x-h)^2 + k$$

$$y = a(x-0)^2 + k$$

$$y = ax^2 + k$$

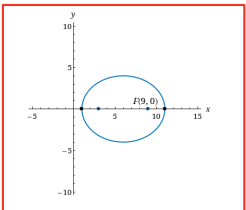
$$0 = a(1)^2 + k$$

$$0 = a + 5$$

$$a = -5$$

$$y = -5x^2 + 5$$

8) Find an equation for the conic whose graph is shown



x-major  
center: (6, 0)

$$2a = 10$$

$$a = 5$$

$$a^2 = 25$$

$$c = 9 - 6$$

$$c = 3$$

$$c^2 = a^2 - b^2$$

$$9 = 25 - b^2$$

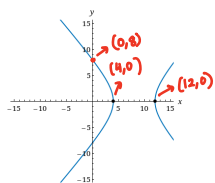
$$b^2 = 25 - 9$$

$$b^2 = 16$$

$$\frac{(x-6)^2}{25} + \frac{(y-0)^2}{16} = 1$$

$$\frac{(x-6)^2}{25} + \frac{y^2}{16} = 1$$

9) Find an equation for the conic whose graph is shown



x-major  
center:  $\left(\frac{4+12}{2}, \frac{0+0}{2}\right)$   
 $(8, 0)$

$$a = 12 - 8$$

$$a = 4$$

$$a^2 = 16$$

$$\frac{(x-8)^2}{16} - \frac{y^2}{b^2} = 1$$

$$\frac{(0-8)^2}{16} - \frac{8^2}{b^2} = 1$$

$$4 - \frac{64}{b^2} = 1$$

$$b^2 = \frac{64}{3}$$

$$\frac{(x-8)^2}{16} - \frac{3y^2}{64} = 1$$

10) Find the standard equation for the conic section with the given properties

The parabola passes through the point  $(7, -2)$  with vertex  $(5, -1)$  and vertical axis of symmetry

$$(x-5)^2 = 4p(y+1)$$

$$(7-5)^2 = 4p(-2+1)$$

$$4 = -4p$$

$$p = -1$$

$$(x-5)^2 = -4(y+1)$$

11) Complete the square to determine whether the graph of the equation is an ellipse, a parabola, a hyperbola, or a degenerate conic. Find appropriate values/equations.

$$x^2 + 4x + 12y + 4 = 0$$

parabola

$$x^2 + 4x = -12y - 4$$

$$x^2 + 4x + 4 = -12y - 4 + 4$$

$$(x+2)^2 = -12y$$

$$\text{vertex: } (-2, 0)$$

$$4p = -12$$

$$p = -3$$

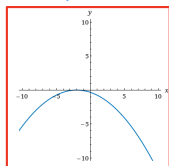
$$\text{Focus: } (-2, 0 + (-3))$$

$$(-2, -3)$$

$$\text{Directrix: } y = 0 - (-3)$$

$$y = 3$$

Graph



12) Complete the square to determine whether the graph of the equation is an ellipse, a parabola, a hyperbola, or a degenerate conic. Find appropriate values/equations.

$$4x^2 + 25y^2 - 40x + 150y + 225 = 0$$

$$4(x^2 - 10x) + 25(y^2 + 6y) = -225$$

$$4(x^2 - 10x + 25) + 25(y^2 + 6y + 9) = -225 + 4(25) + 25(9)$$

$$4(x-5)^2 + 25(y+3)^2 = 100$$

$$\frac{(x-5)^2}{25} + \frac{(y+3)^2}{4} = 1$$

ellipse

Center:  $(5, -3)$

$$c^2 = a^2 - b^2$$

$$c^2 = 25 - 4$$

$$c^2 = 21$$

$$c = \sqrt{21}$$

Focus:  $(5 - \sqrt{21}, -3)$

Focus:  $(5 + \sqrt{21}, -3)$

Vertex:  $(5-5, -3)$

$(0, -3)$

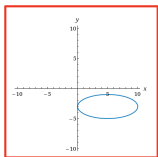
Vertex:  $(5+5, -3)$

$(10, -3)$

length of major axis:  $2a = 2 \cdot 5 = 10 \text{ un}$

length of minor axis:  $2b = 2 \cdot 2 = 4 \text{ un}$

Graph



13) Complete the square to determine whether the graph of the equation is an ellipse, a parabola, a hyperbola, or a degenerate conic. Find appropriate values/equations.

$$x^2 + 81 = 9(y^2 + 2x)$$

$$x^2 + 81 = 9y^2 + 18x$$

$$x^2 - 18x - 9y^2 = -81$$

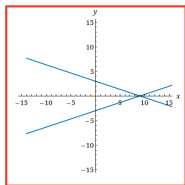
$$(x^2 - 18x + 81) - 9y^2 = -81 + 81$$

$$(x-9)^2 - 9y^2 = 0$$

$$(x-9-3y)(x-9+3y) = 0$$

degenerate conic

Graph



14) Complete the square to determine whether the graph of the equation is an ellipse, a parabola, a hyperbola, or a degenerate conic. Find appropriate values/equations.

$$x^2 - y^2 = 2(x-y) + 1$$

$$x^2 - y^2 = 2x - 2y + 1$$

$$x^2 - 2x - y^2 + 2y = 1$$

$$(x^2 - 2x + 1) - (y^2 - 2y + 1) = 1 + 1 - 1$$

$$(x-1)^2 - (y-1)^2 = 1$$

hyperbola

Center:  $(1, 1)$

$$a^2 = 1$$

$$b^2 = 1$$

$$c^2 = a^2 + b^2$$

$$c^2 = 1 + 1$$

$$c^2 = 2$$

$$c = \sqrt{2}$$

Focus:  $(1 - \sqrt{2}, 1)$

Focus:  $(1 + \sqrt{2}, 1)$

Vertex:  $(1 - 1, 1)$

$(0, 1)$

Vertex:  $(1 + 1, 1)$

$(2, 1)$

Asymptotes:  $y - 1 = \pm \frac{1}{1} (x - 1)$

$$y - 1 = \pm (x - 1)$$

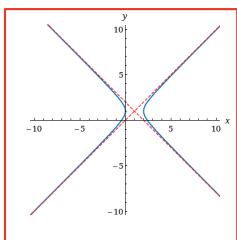
$$y - 1 = x - 1$$

$$y = x$$

$$y - 1 = -(x - 1)$$

$$y = -x + 2$$

Graph



- 15) Consider a conic section with the following properties  
ellipse with the center  $C(3, -1)$ , vertices  $V_1(-7, -1)$  and  $V_2(13, -1)$ , and foci  $F_1(-3, -1)$  and  $F_2(9, -1)$   
Determine the values of  $h$ ,  $k$ ,  $a$ , and  $b$ .

$$h = 3$$

$$k = -1$$

$$a = |13 - 3| = 10$$

$$a = 10$$

$$c = |9 - 3| = 6$$

$$b^2 = a^2 - c^2$$

$$b^2 = 10^2 - 6^2$$

$$b^2 = 64$$

$$b = 8$$

Find the standard equation for the conic section with the given properties.

$$\frac{(x-3)^2}{100} + \frac{(y+1)^2}{64} = 1$$

- 16) Find the standard equation for the conic section with the given properties.

ellipse with the center  $C(1, -5)$ , vertices  $V_1(-9, -5)$  and  $V_2(11, -5)$ , and foci  $F_1(-5, -5)$  and  $F_2(7, -5)$

$$a = |11 - 1|$$

$$a = 10$$

$$a^2 = 100$$

$$c = |7 - 1| = 6$$

$$b^2 = a^2 - c^2$$

$$b^2 = 10^2 - 6^2$$

$$b^2 = 100 - 36$$

$$b^2 = 64$$

$$\frac{(x-1)^2}{100} + \frac{(y+5)^2}{64} = 1$$

- 17) Find the standard equation for the conic section with the given properties.  
hyperbola with vertices  $V_1(-2, -2)$  and  $V_2(4, -2)$  and foci  $F_1(-5, -2)$  and  $F_2(7, -2)$   
Center:  $\left(\frac{-2+4}{2}, \frac{-2+(-2)}{2}\right)$   
 $(1, -2)$

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$a = 4 - 1 = 3$$

$$a^2 = 9$$

$$c = 7 - 1 = 6$$

$$c^2 = 36$$

$$c^2 = a^2 + b^2$$

$$36 = 9 + b^2$$

$$b^2 = 27$$

$$\frac{(x-1)^2}{9} - \frac{(y+2)^2}{27} = 1$$

- 18) Consider a conic section with the following properties  
parabola with vertex  $(-4, 7)$  and directrix  $y = 5$   
Determine the values of  $h, k$  and  $p$ .

$$h = -4$$

$$k = 7$$

$$p = 7 - 5 = 2$$

$$p = 2$$

Find the standard equation for the conic section with the given properties.

$$(x - (-4))^2 = 4(2)(y - 7)$$

$$(x + 4)^2 = 8(y - 7)$$

- 19) Consider a conic section with the following properties.  
the hyperbola with foci  $F_1(2, -15)$  and  $F_2(2, 15)$  that passes through the point  $(2, 12)$

Determine the values of  $h, k, a$ , and  $b$ .

$$h = \frac{2+2}{2}$$

$$h = 2$$

$$k = \frac{-15+15}{2}$$

$$k = 0$$

$$c = |15 - 0| = 15$$

$$\text{distance from } P(2, 12) \text{ to } F_1(2, -15): \sqrt{(2-2)^2 + (12-(-15))^2}$$

$$\text{distance from } P(2, 12) \text{ to } F_2(2, 15): \sqrt{(2-2)^2 + (12-15)^2} = 3$$

$$2a = |27 - 3| = 24$$

$$a = 12$$

$$a^2 = 144$$

$$a = 12$$

$$c^2 = a^2 + b^2$$

$$15^2 = 144 + b^2$$

$$b^2 = 81$$

$$b = 9$$

$$\frac{(y-0)^2}{144} - \frac{(x-2)^2}{81} = 1$$

- 20) Find the standard equation for the conic section with the given properties.  
parabola that passes through point  $(5, 1)$ , with vertex  $V(-1, 2)$   
and horizontal axis of symmetry

$$(y-k)^2 = 4p(x-h)$$

$$(y-2)^2 = 4p(x+1)$$

$$(1-2)^2 = 4p(5+1)$$

$$1 = 24p$$

$$4p = \frac{1}{6}$$

$$(y-2)^2 = \frac{1}{6}(x+1)$$